Learning Objectives

Upon completion of this chapter, students will be able to

- calculate the linear approximation of a function of one variable at a point
- approximate the value of a function using its linear approximation without calculators
- calculate the differential of a function
- apply linear approximation, differential in problems from real life situations
- find partial derivatives of a function of more than one variable
- calculate the linear approximation of a function of two or more variables
- determine if a given function of several variables is homogeneous or not
- apply Euler's theorem for homogeneous functions.

8.2 Linear Approximation and Differentials

8.2.1 Linear Approximation

In this section, we introduce linear approximation of a function at a point. Using the linear approximation, we shall estimate the function value near a chosen point. Then we shall introduce differential of a real-valued function of one variable, which is also useful in applications.

Let $f:(a,b)\to\mathbb{R}$ be a differentiable function and $x\in(a,b)$. Since f is differentiable at x, we have

$$\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = f'(x) \qquad \ldots (1)$$

If Δx is small, then by (1) we have

$$f(x+\Delta x)-f(x) \approx f'(x)\Delta x\,; \qquad \ldots (2)$$

which is equivalent to

$$f(x+\Delta x) \approx f(x)+f'(x)\Delta x, \qquad \ldots (3)$$

where $\approx$ means **"approximately"** equal. Also, observe that as the independent variable changes from x to $x+\Delta x$, the function value changes from $f(x)$ to $f(x+\Delta x)$. Hence if Δx is small and the change in the output is denoted by Δf or Δy, then (2) can be rewritten as

$$\text{change in the output} = \Delta y = \Delta f = f(x+\Delta x)-f(x) \approx f'(x)\Delta x\,.$$

Note that (3) helps in approximating the value of $f(x+\Delta x)$ using $f(x)$ and $f'(x)\Delta x$. Also, for a fixed x_0, $y(x)=f(x_0)+f'(x_0)(x-x_0), x\in\mathbb{R}$, gives the tangent line for the graph of f at $(x_0,f(x_0))$ which gives a good approximation to the function f near x_0. This leads us to define

Definition 8.1 (Linear Approximation)

Let $f:(a,b)\to\mathbb{R}$ be a differentiable function and $x_0\in(a,b)$. We define the linear approximation L of f at x_0 by

$$L(x) = f(x_0)+f'(x_0)(x-x_0), \quad \forall x\in(a,b) \qquad \ldots (4)$$

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